

MLP VS CNN FOR FASHION-MNIST IMAGE CLASSIFICATION

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PROBLEM AND DATASET

Problem: Classify clothing images into 10 categories

Fashion-MNIST Dataset:

- 60,000 training images, 10,000 test images
- 28×28 grayscale pixels
- 10 classes: T-shirt/top, Trouser, Pullover, Dress, Coat, Sandal, Shirt, Sneaker, Bag, Ankle boot



METHODS: MLP VS CNN

MLP (Baseline):

- Input 784 pixels $\rightarrow$ 128 neurons $\rightarrow$ 10 classes
- Parameters: $\sim 101,000$

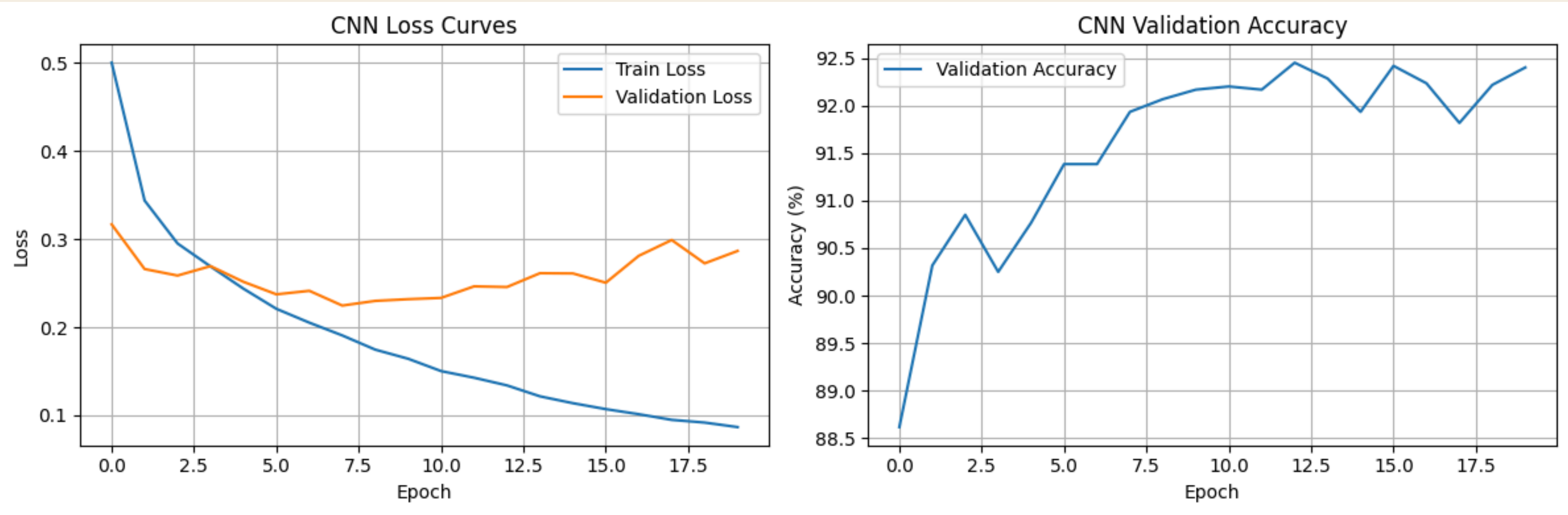
CNN (Deep Learning):

- Conv1 (1 $\rightarrow$ 32) + BatchNorm + MaxPool
- Conv2 (32 $\rightarrow$ 64) + BatchNorm + MaxPool
- FC (3136 $\rightarrow$ 256) + Dropout(0.5) $\rightarrow$ 10 classes
- Parameters: 824,650

Why CNN is better: Preserves spatial structure, extracts edges and textures

RESULTS

Model	Accuracy	F1-Score
MLP	88.44%	0.8832
CNN	92.98%	0.9298

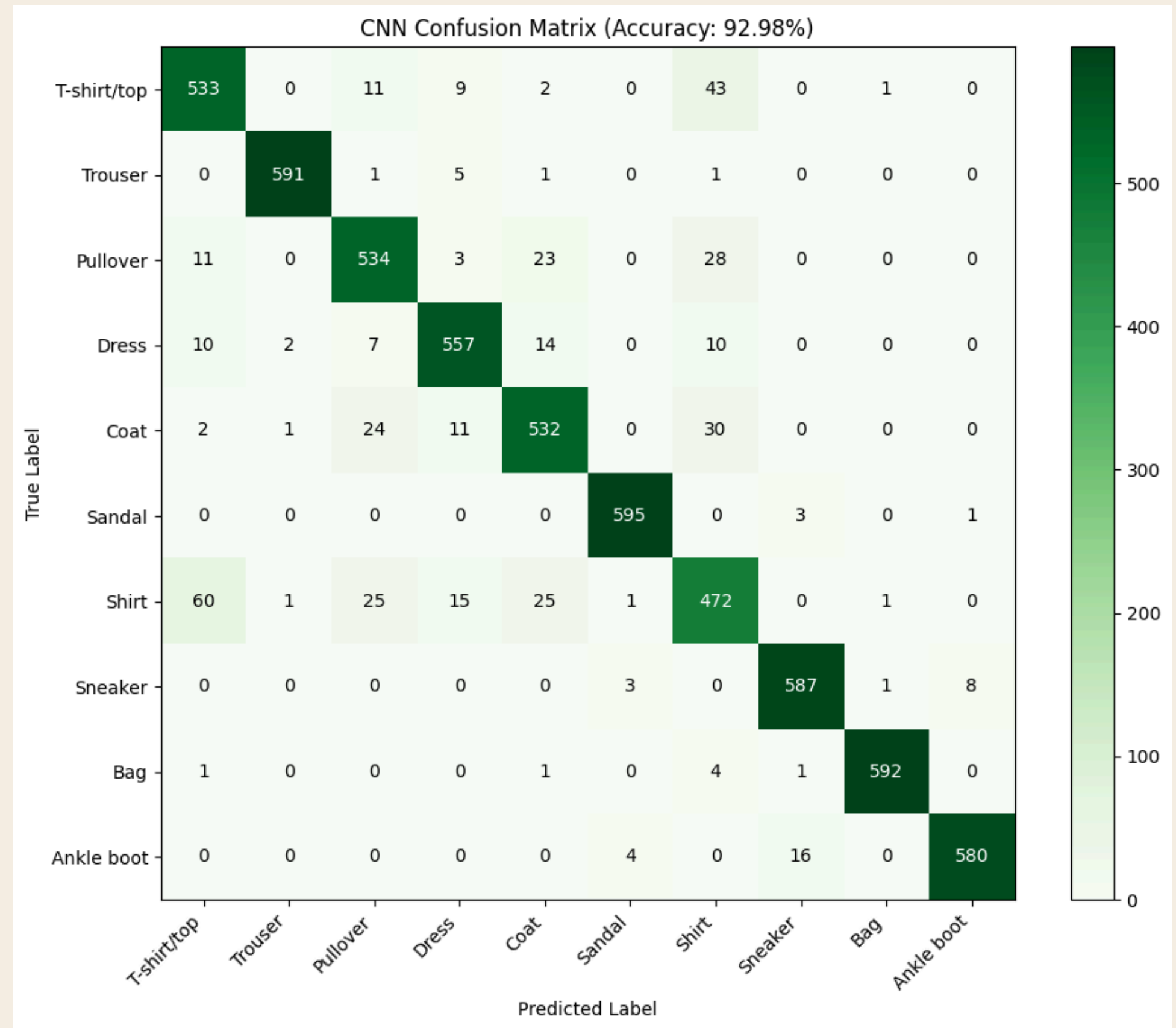


ERROR ANALYSIS

Top-5 CNN Mistakes:

1. Dress → Coat (60 times)
2. Shirt → T-shirt/top (~40 times)
3. Shirt → Coat (~35 times)
4. Pullover → Coat (~30 times)
5. T-shirt/top → Shirt (~25 times)

Reason: Visually similar clothing items



CONCLUSION

Achievements:

- CNN: 92.98% accuracy
- MLP: 88.44% accuracy
- Improvement: +4.54%

Future improvements:

- Data augmentation (rotation, shift, zoom)
- Pre-trained models (ResNet-18)



THANK YOU FOR YOUR ATTENTION